SOC L1 LAB · INC-006

Reproduction Guide: INC-006 Lab

Creating an approved change, tripping the rules, and closing the case with evidence

INCIDENT	INC-006, Authorized change triage
REPORT DATE	2026-11-01
CLASSIFICATION	Lab exercise, sanitized for publication
ENVIRONMENT	Wazuh 4.14.7 · Debian 12 · QEMU/KVM
SEVERITY	None (authorized change)

Independent portfolio lab. No production systems, client data, or real credentials.

Contents

1. Prerequisites
2. Part A: Build the change package
3. Part B: Deploy
4. Part C: Collect the alerts
5. Part D: Triage
6. Part E: Close
7. Part F: Rollback (optional)
8. Troubleshooting
9. Checklist

This guide reproduces the authorized-change triage on the existing lab. It assumes the Wazuh stack, the Debian endpoint, and rules 100129/100130 from INC-004 are running.

Time

15–20 minutes. **Result**

Four persistence alerts, a matching change record, a written closure, and a rollback path.

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1. Prerequisites

Requirement	Check
Wazuh stack	`docker compose ps`
Agent active	`agent_control -l` shows `001 soc-victim Active`
Rules 100129/100130	`grep -c` in the container
FIM realtime on `/etc`	`grep -n 'realtime.*etc'`
Admin sudo	`analyst` can run `sudo -n true`

```
export LAB="$HOME/Documents/Eterna/soc-l1-lab"
export VMKEY="$LAB/lab/vm/keys/id_ed25519"
export VM="analyst@127.0.0.1"
export VMPORT=2222
```

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2. Part A: Build the change package

Create the files locally, compute hashes, and write the change record before deploying.

```
mkdir -p /tmp/inc006
cat > /tmp/inc006/log-maintenance <<'EOF'
# Nightly cleanup of rotated logs older than 7 days
15 2 * * * root /usr/bin/find /var/log -type f -name '*.gz' -mtime +7 -delete
EOF

cat > /tmp/inc006/logrotate-check.service <<'EOF'
[Unit]
Description=Logrotate health check

[Service]
Type=oneshot
ExecStart=/usr/bin/true
EOF
```

```
cat > /tmp/inc006/logrotate-check.timer <<'EOF'
[Unit]
Description=Daily logrotate health check

[Timer]
OnCalendar=daily
Persistent=true
Unit=logrotate-check.service

[Install]
WantedBy=timers.target
EOF

sha256sum /tmp/inc006/log-maintenance \
          /tmp/inc006/logrotate-check.service \
          /tmp/inc006/logrotate-check.timer
```

Write the change record with the hashes, the approver, the window, and the rollback. A template is in `04-change-setup.md`.

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3. Part B: Deploy

```
scp -q -i "$VMKEY" -P "$VMPORT" \
  /tmp/inc006/CHG-2026-1101-01.md \
  /tmp/inc006/log-maintenance \
  /tmp/inc006/logrotate-check.service \
  /tmp/inc006/logrotate-check.timer \
  "$VM":/tmp/

ssh -i "$VMKEY" -p "$VMPORT" "$VM" '
mkdir -p ~/changes
cp /tmp/CHG-2026-1101-01.md ~/changes/
sudo cp /tmp/log-maintenance /etc/cron.d/log-maintenance
sudo cp /tmp/logrotate-check.service /etc/systemd/system/logrotate-check.service
sudo cp /tmp/logrotate-check.timer /etc/systemd/system/logrotate-check.timer
sudo chmod 644 /etc/cron.d/log-maintenance
sudo systemctl daemon-reload
sudo systemctl enable --now logrotate-check.timer
systemctl is-active logrotate-check.timer
sudo sha256sum /etc/cron.d/log-maintenance \
  /etc/systemd/system/logrotate-check.service \
  /etc/systemd/system/logrotate-check.timer'
```

Expected:

- timer `active`.
- three hashes matching the change record.

- an enable symlink under `/etc/systemd/system/timers.target.wants/`.

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4. Part C: Collect the alerts

```
sleep 8
docker exec single-node-wazuh.manager-1 sh -c \
  'grep -E "\"id\":\"100129\"|\"id\":\"100130\"" /var/ossec/logs/alerts/alerts.json | tail -6' \
  | jq -r '[.timestamp,.rule.id,.syscheck.path] | @tsv'
```

Expected:

```
100129  /etc/cron.d/log-maintenance
100130  /etc/systemd/system/logrotate-check.service
100130  /etc/systemd/system/logrotate-check.timer
100130  /etc/systemd/system/timers.target.wants/logrotate-check.timer
```

There may be duplicate or batched events. The four paths above are the signal.

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5. Part D: Triage

5.1 Authorization

```
ssh -i "$VMKEY" -p "$VMPORT" "$VM" 'cat ~/changes/CHG-2026-1101-01.md'
```

Check:

- status is Approved.
- the requester and approver are named.
- the window includes the deployment time.
- the file list matches what appeared.
- the hashes match.

5.2 Integrity

```
ssh -i "$VMKEY" -p "$VMPORT" "$VM" '
sudo sha256sum /etc/cron.d/log-maintenance \
  /etc/systemd/system/logrotate-check.service \
  /etc/systemd/system/logrotate-check.timer'
```

Compare every hash to the record. A mismatch is an escalation, not a closure.

5.3 Behaviour

```
ssh -i "$VMKEY" -p "$VMPORT" "$VM" '
cat /etc/cron.d/log-maintenance
systemctl cat logrotate-check.service
systemctl cat logrotate-check.timer
systemctl list-timers logrotate-check.timer --no-pager'
```

Check that the contents are what the change says: a log cleanup and a no-op health check. Look for an unexpected `ExecStart`, a network call, or a new account.

5.4 Follow-on activity

```
ssh -i "$VMKEY" -p "$VMPORT" "$VM" \
  'sudo journalctl --since "2026-10-31 23:35:45" --until "2026-10-31 23:37:00" \
   --no-pager | grep -Ei "failed password|invalid user|authentication failure"'
```

Expected: no output.

Also check FIM/audit for unexpected files or processes in the same window.

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6. Part E: Close

Write the ticket with:

- the change ID and approver.
- the four alert paths.
- the hash comparison result.
- the file contents summary.
- the absence of follow-on activity.
- the decision: authorized change, no incident.
- the note that this is not a false positive.

No containment. No rule change. Keep the detection enabled.

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7. Part F: Rollback (optional)

The change is a lab artifact. To remove it:

```
ssh -i "$VMKEY" -p "$VMPORT" "$VM" '
sudo systemctl disable --now logrotate-check.timer
sudo rm -f /etc/systemd/system/logrotate-check.service \
           /etc/systemd/system/logrotate-check.timer
sudo systemctl daemon-reload
```

```
sudo rm -f /etc/cron.d/log-maintenance
systemctl list-timers --all | grep -i logrotate || echo "timer gone"'
```

If you roll back, record the rollback in the change record. Do not remove it silently.

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8. Troubleshooting

Symptom	Cause	Fix
No 100129	`/etc` not FIM realtime	Verify syscheck. Restart agent
No 100130	Unit written outside `/etc/systemd/system`	Use the recorded path
Hash mismatch	File changed after the record	Stop. Escalate. Do not close
Timer inactive	`enable --now` not run	`systemctl enable --now logrotate-check.timer`
Timer runs immediately	`OnCalendar=daily` semantics	Check `systemctl list-timers`. Adjust if needed
Alert missing in Wazuh	Manager queue/filter	Check agent and manager logs
Legit change looks like an incident	No change record or no hashes	Escalate. The evidence is missing

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9. Checklist

- [ ] Agent active
- [ ] Rules 100129/100130 loaded
- [ ] Change record written with hashes before deploy
- [ ] Cron file, service, timer deployed
- [ ] Timer active
- [ ] Hashes match the record
- [ ] Four alert paths collected
- [ ] No authentication/process anomalies
- [ ] Decision documented as authorized change
- [ ] Rollback path recorded